

3.3 App Components and Key Features

This subsection covers interface layout, on-device ML Kit inference, and local storage, using the same terms as the codebase (Flutter/Dart, Digital Ink, *flutter_tts*, *shared_preferences*). Figure 1 outlines the practice-time pipeline.



Figure 1. On-device practice pipeline (HC = high-contrast canvas; TTS = text-to-speech). Arrows: main control/data flow in one session.

Frontend (presentation layer)

- **High-contrast presentation and enlarged typography.** Dark scaffold, large base text, and global linear text scaling support low vision; long-press on the canvas toggles a higher-contrast drawing view.

- **Navigation and task routing.** Home splits work into lowercase, uppercase, and numerals (grid then shared workspace). *Poem Practice* appears when the UI language is English (short phrases and longer verses).

- **Voice-guided writing workspace.** Template plus digital-ink canvas; TTS (Cantonese or English) can differ from Chinese/English on-screen labels. Double-tap replay, pinch zoom, brush/eraser; glyph drills add round presets and a progress bar.

- **Passage-oriented (poem) composition mode.** Scrollable passage; one active grapheme at a time while speech tracks the current word and character.

- **Landscape layout and orientation control (poem practice only).** Only poem practice uses a portrait/landscape split via *OrientationBuilder*: portrait stacks word, progress, and canvas; landscape uses a ~100px left strip (word, progress, circular speak/clear/next) beside a full-height wide canvas. Landscape follows physical rotation or an app-bar lock using *SystemChrome.setPreferredOrientations* (landscape left/right vs portrait-up); dispose restores portrait-up. Single-glyph practice has no such split.

- **Terminal analytics for passage tasks.** After the last token, TTS and a *Poem Summary* dialog report accuracy, mean time per character, session duration, and difficult words.

- **Multilingual coverage.** Latin upper/lowercase, Arabic numerals, Traditional Chinese numerals (1-10 in the grid); bilingual UI with Cantonese or English TTS.

- **Accessibility metadata.** Semantics describe canvas actions (replay, high contrast, zoom) for screen readers.

Backend and local data handling (client-only)

Client-only: no login, no cloud ink API, no Firebase. ML Kit scores on-device. *shared_preferences* holds UI/voice locales and JSON score lists per practice key (not synced). High-contrast canvas is session state, not a saved theme.

3.4 Core Features

(1) Voice-guided handwriting practice. Spoken steps scaffold graphemes, words, and short lines. UI language and TTS are independent. On-device scoring drives spoken percentages and, in poem mode, auto-advance or retry prompts.

(2) Poem practice with continuous writing and structured feedback. Same ML Kit path as drills inside English passages; TTS follows the active span; scrollable text keeps context. End of session combines speech and summary UI (accuracy, timing, weak words).

(3) Multilingual script support. Latin cases, Arabic digits, Traditional Chinese numerals, bilingual UI, flexible TTS; custom fonts and accessibility features keep tasks readable with all inference local.